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# Ocular drift transforms retinal image statistics and spatiotemporal receptive fields

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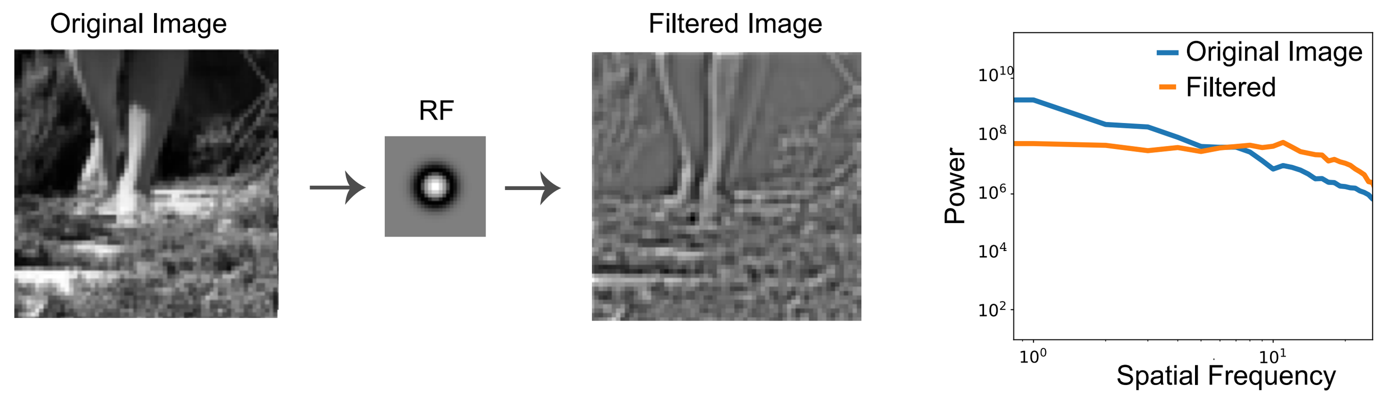
## Background: Efficient Coding

- Efficient coding states that sensory neurons should adapt their coding strategies to **maximize information** about natural stimuli given biophysical and metabolic constraints
- It predicts that sensory representations match the structure of the environment, such as whitening correlated input in natural images [6,7]
- This framework explains features of sensory systems (e.g., center surround receptive fields remove redundancy in natural scenes) and can be learned from data [1,4]

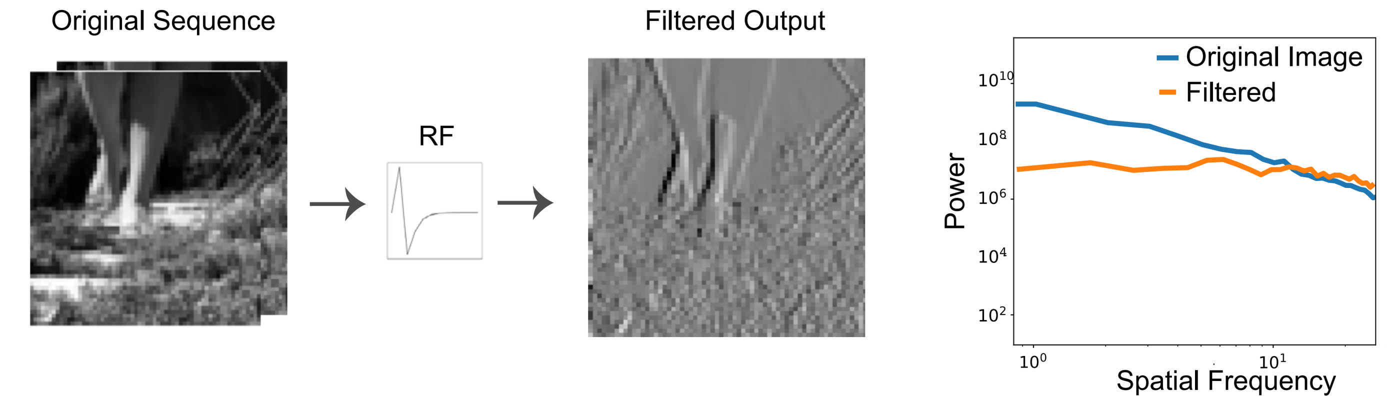
## Whitening: two competing explanations

**Spatial RFs**: whitening results from center surround receptive fields




- Center surround receptive fields remove low frequencies, effectively flattening the power spectrum [5]

**Temporal**: whitening results from temporal receptive fields and spatial shifts from eye movements



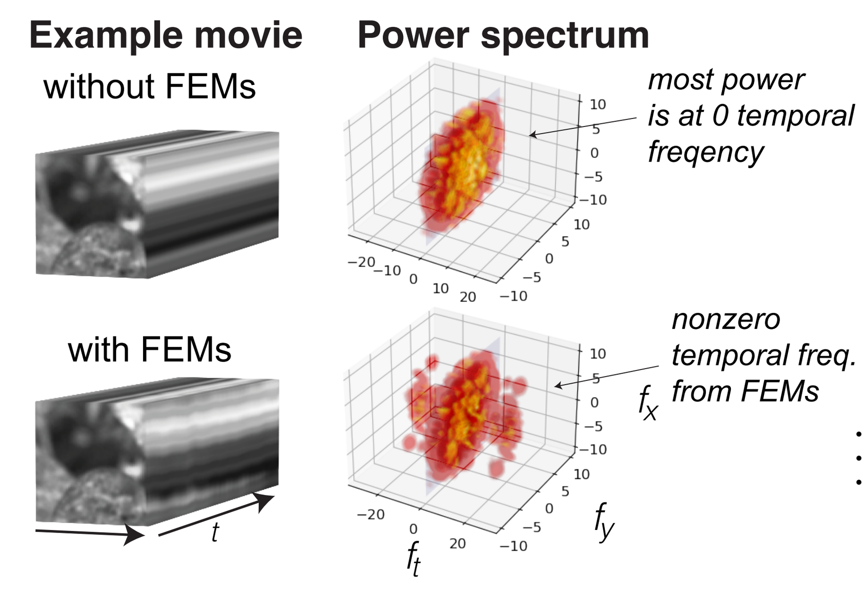

- Spatial displacements from fixational eye movements shift spatial power into temporal modulations, resulting in a whitened spatial power spectrum [8]

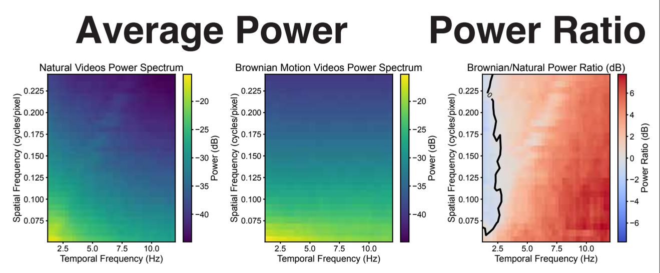
## Eye movements are part of the natural statistics

- Previous work shows that human fixational eye movements follow a temporal whitening strategy consistent with the example shown above [3,10]
- Because real visual input always includes eye movements, we ask: if fixational drift whitens the input, how does this affect optimal encoding strategies and learned representations in efficient coding models?
- To test this, we train an information-maximization encoder on movies with and without simulated fixational eye movements and compare the learned representations

## Generating natural movies with ocular drift

Natural movies resampled with Brownian motion redistribute power




*Eye movements shift power to higher temporal frequencies*

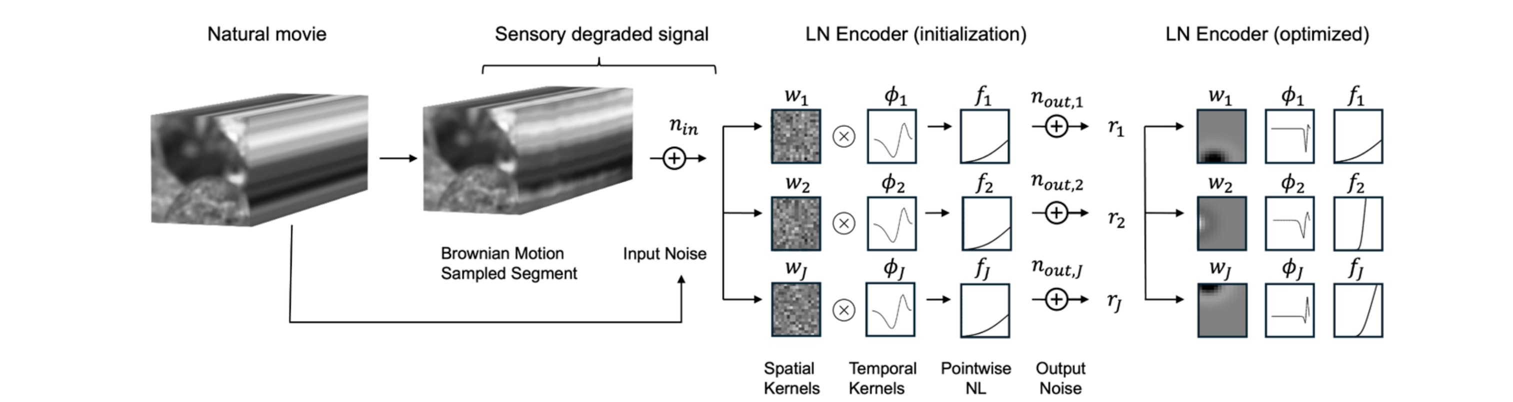
Movies from the Chicago motion Database [2]

### Simulated Drift

- Random walk in pixel space
- At each time step $t$, sampling position coordinates
- $x(t)$ and $y(t)$ are displaced by a Gaussian perturbation
  - $x(t) = x(t-1) + \sqrt{2D\Delta t} \cdot \Delta X$
  - $y(t) = y(t-1) + \sqrt{2D\Delta t} \cdot \Delta Y$
    - $D$ is a diffusion constant ($40\ \mathrm{arcmin}^2/\mathrm{s}$)
    - $\Delta t$ is the frame rate of 30 fps
    - $\Delta X$ and $\Delta Y$ are randomly drawn from a normal distribution $\sim \mathcal{N}(0,1)$

## Space-time separable encoding model

We fit a space-time separable linear-nonlinear encoder




Generator Signal

$$y_j = \mathbf{w}_j^\top(\mathbf{x} + \mathbf{n}_x)$$

Unit norm weight vector of neuron; Input signal; Gaussian noise (input/sensory noise)

Neural Response

$$r_j = f_j(y_j) + n_r$$

point-wise NL; Generator Signal; Gaussian noise (output/neural noise)

Scalar output

- Model architecture following [1]
- Space-time separable filters
- Temporal filter is parameterized as a difference of exponentials
- Spatial filter is non-parametric (random initialization)
- Learned Softplus nonlinearity

## Model objective: Infomax

Gradient descent on the Infomax loss

- **Goal:** maximize information transfer between input $\mathbf{x} \in X$, and the neural response $\mathbf{r} \in R$, subject to metabolic cost of firing spikes

$$\underbrace{I(X;R)}_{\text{Mutual Information}} - \sum_j \underbrace{\lambda_j \langle r_j \rangle}_{\text{Metabolic Cost}}$$

Input; Neural Response; Firing Rate

- To make this computationally tractable, mutual information is expressed as the conditional differential entropy of a multivariate Gaussian

$$\text{maximize} \quad \log \frac{\det\left(\mathbf{G}\mathbf{W}^\top(\mathbf{C}_x + \mathbf{C}_{n_{\mathrm{in}}})\mathbf{W}\mathbf{G} + \mathbf{C}_{n_{\mathrm{out}}}\right)}{\det(\mathbf{G}\mathbf{W}^\top\mathbf{C}_{n_{\mathrm{in}}}\mathbf{W}\mathbf{G} + \mathbf{C}_{n_{\mathrm{out}}})}$$

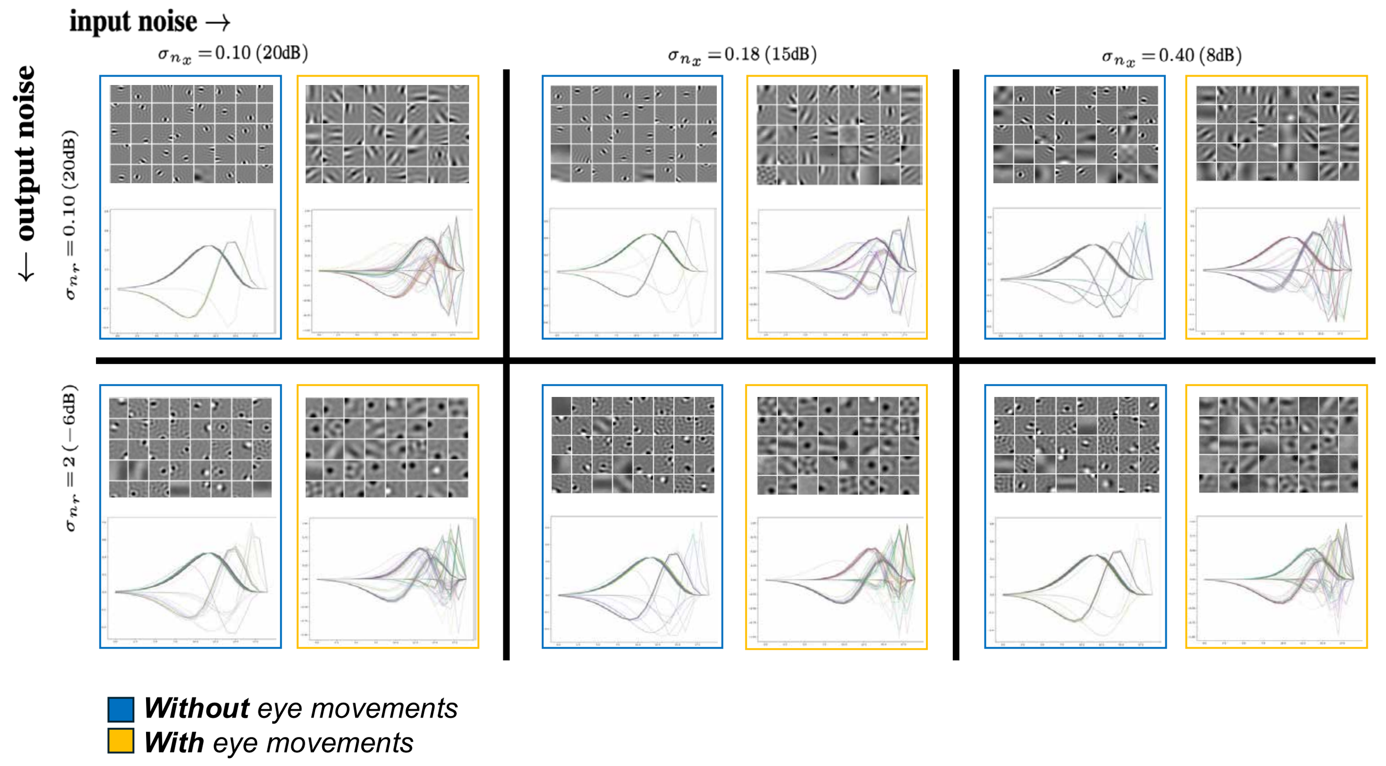
- The model will learn filters that maximize information about the input while minimizing cost, enforced by constraining the mean output of each neuron

$$\text{subject to} \quad \mathbb{E}[r_j] = 1.$$

- Infomax loss using local analytic Gaussian approximations from [1,4]
- Constrained by a metabolic cost term (implemented as a Lagrange multiplier)
- All parameters fit using stochastic gradient descent with the Adam optimizer

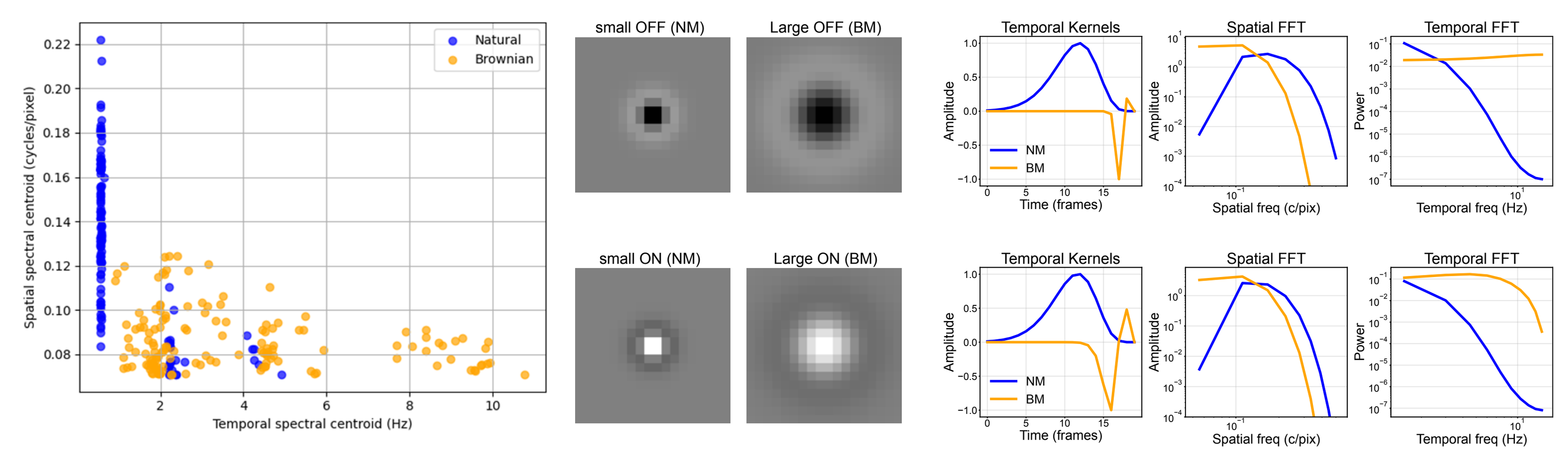
## Results: eye movements alter learned representations

Spatial RFs become more diffuse and temporal RFs become more biphasic when trained on movies jittered with Brownian motion (eye movements)




## Learned weights shift to temporal processing

Filters trained on Brownian-jittered movies tile higher temporal frequencies and lower spatial frequencies




## Conclusions and future work

- We used the Infomax objective to train a simple LN encoder on natural movies and those same movies resampled by simulated fixational drift
- Encoder neurons trained in the presence of drift were more **temporally bandpass** and more **spatially lowpass** than the encoders trained on movies without drift
- Eye movements modify the distribution of natural statistics and alter the encoding schemes available to retinal neurons
- Future work will explore whether our results reflect receptive field trends observed in studies that include eye movements [3,9,10]

## References

[1] Jun et al., 2022
[2] Salisbury and Palmer, 2016
[3] Kuang et al., 2012
[4] Karklin and Simoncelli, 2011
[5] Simoncelli and Olshaisen, 2001
[6] Olshausen & Field, 1996
[7] Karamanlis, 2022
[8] Anderson et al., 2020
[9] Casile et al., 2019
[10] Rucci & Victor, 2015

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